



OHT Brake Pressure - Leakage Test

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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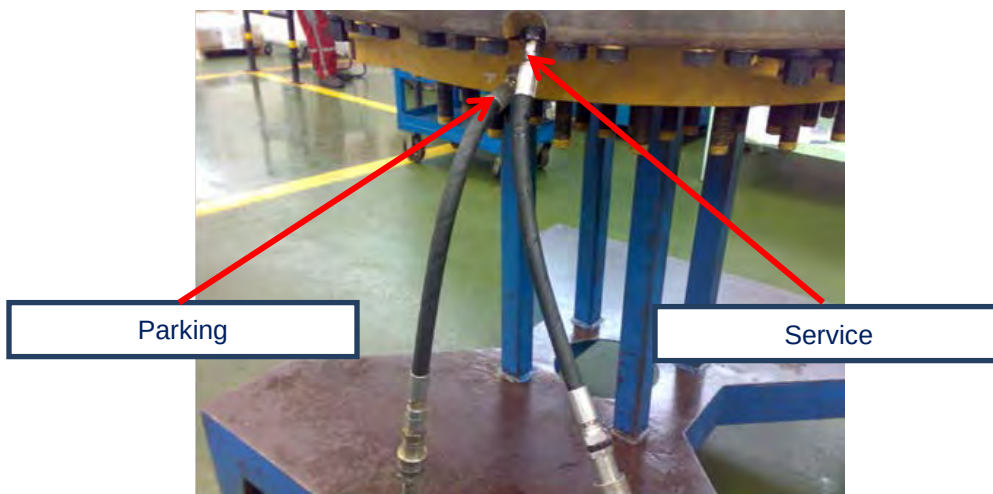
1.0 Introduction

Brake operation is very important and any problem in the assembly can potentially lead to braking issues in the field and at the Finning Chile (CRC Antofagasta) before installing the brake group on the spindle, pressure tests are performed on the service and park brake release systems for leakage.

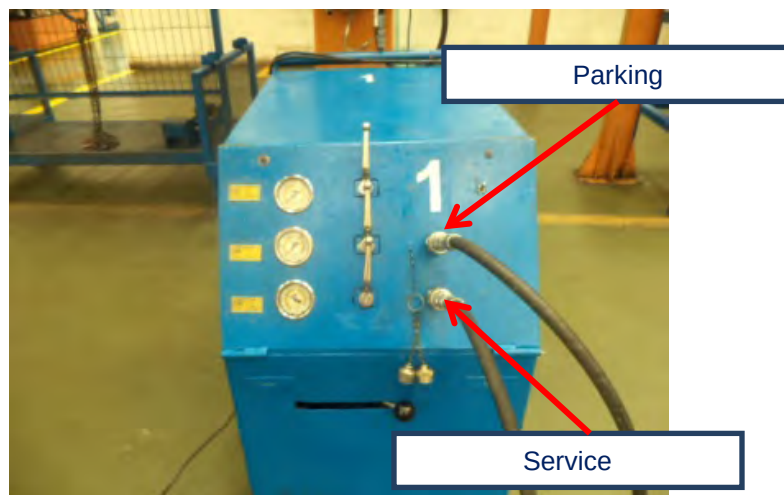
One hydraulic pump is used, to supply regulated oil pressure to the service and park brake oil supply ports.

2.0 Best Practice Description

2.1 Install hoses in the parking and service brake ports.



2.2 Connect same hoses to the test tooling with quick couplings.



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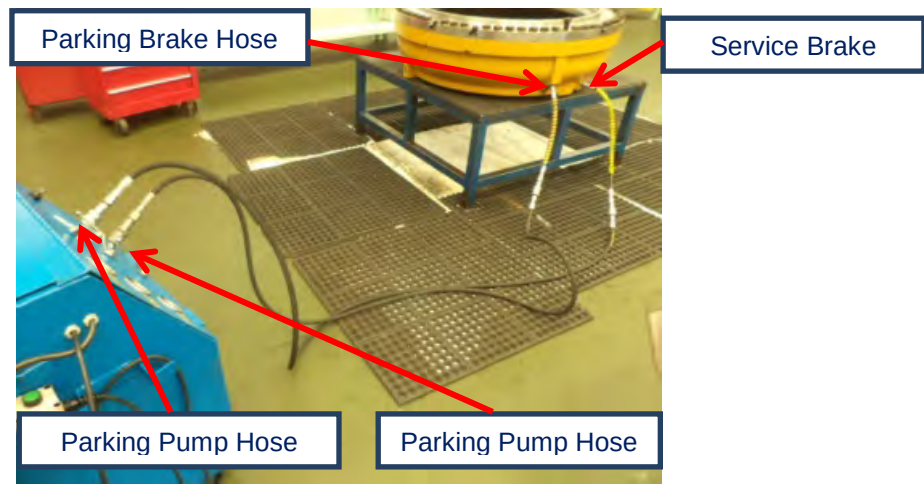
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2.3 Caption shows hose connections between brake group ports and test tooling.

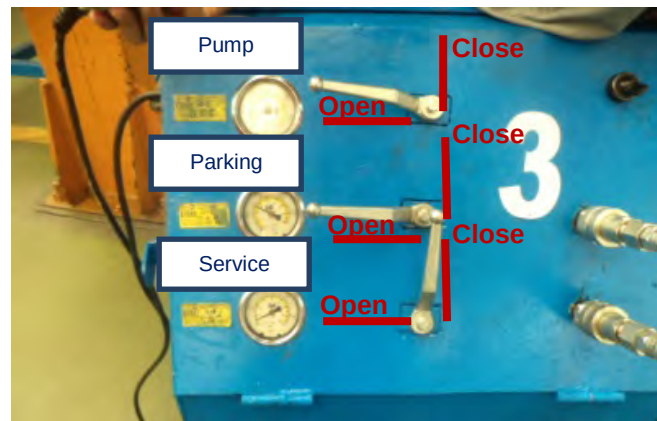


2.4 Apply oil pressure according to the Caterpillar publications for the models been tested.

Model	Document	Parking Pressure	Service Pressure
773	SSNR2986	460+-32 PSI	460+-32 PSI
777	REN3324-SSNR2986	460+-32 PSI	460+-32 PSI
777F	KENR9267	720+-100 PSI	720+-100 PSI
789	SSNR6331	680+-30 PSI	680+-30 PSI
793	SENR6343-SSNR1454-KENR8738-REN8330	680+-30 PSI	680+-30 PSI
797	RSNR2611	728+-20 PSI	728+-20 PSI
797B	RSNR2581	728+-20 PSI	728+-20 PSI
797F	KENR8374	728+-20 PSI	728+-20 PSI

Note: Any time you perform the pressure tests, always check SIS for the latest Information.

2.5 To test the parking brake piston movement, open the pump valve and parking release valve. Then slowly apply pressure, according to the Caterpillar specifications.



Tank (return)

Oil supply

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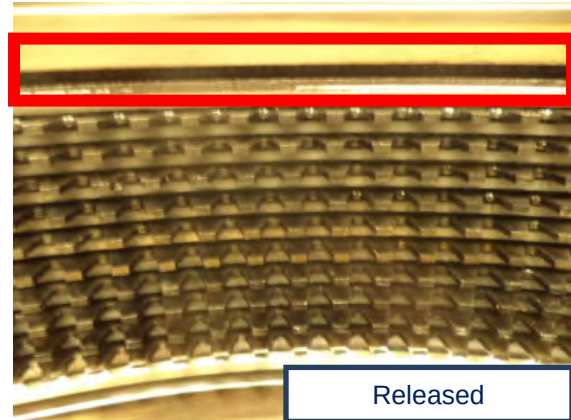
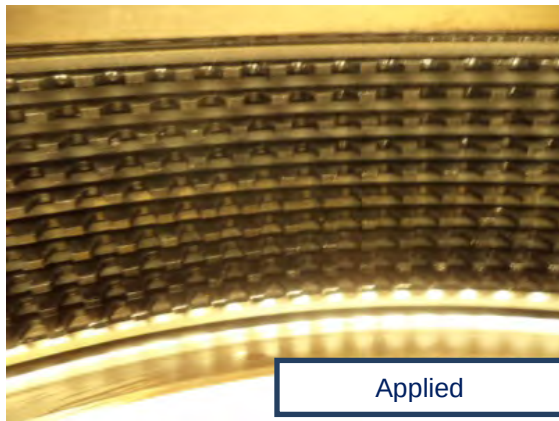
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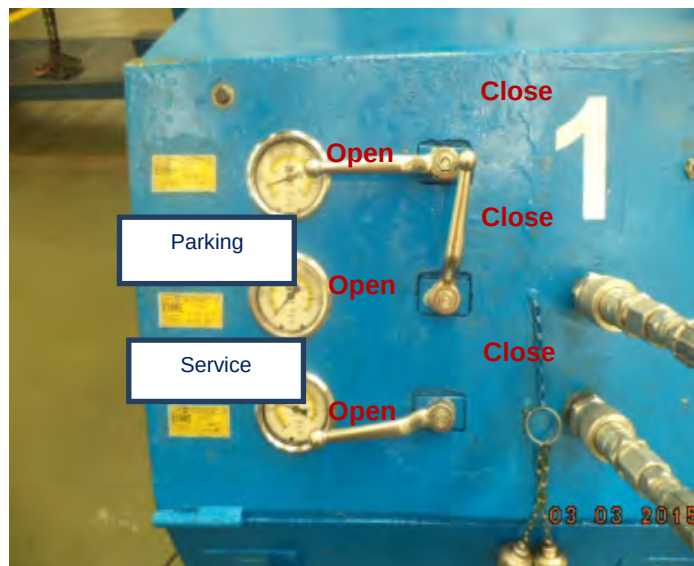
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2.6 When the specified pressure has been reached, check if the parking brake piston, discs and plates have moved down (released) and then applied when the pressure is released as illustrated in the caption.



2.7 Apply and release parking brake pressure (steps 2.5 - 2.6) again and repeat this sequence at least 5 times. In order to ensure proper seating of the piston seals and check for any leakage.

2.8 To check for leakage, leave the parking brake pressure applied (close off parking brake valve at the test tooling) Then open the service brake valve on the test tooling and slowly apply pressure to the service brake piston port per the Caterpillar specifications.



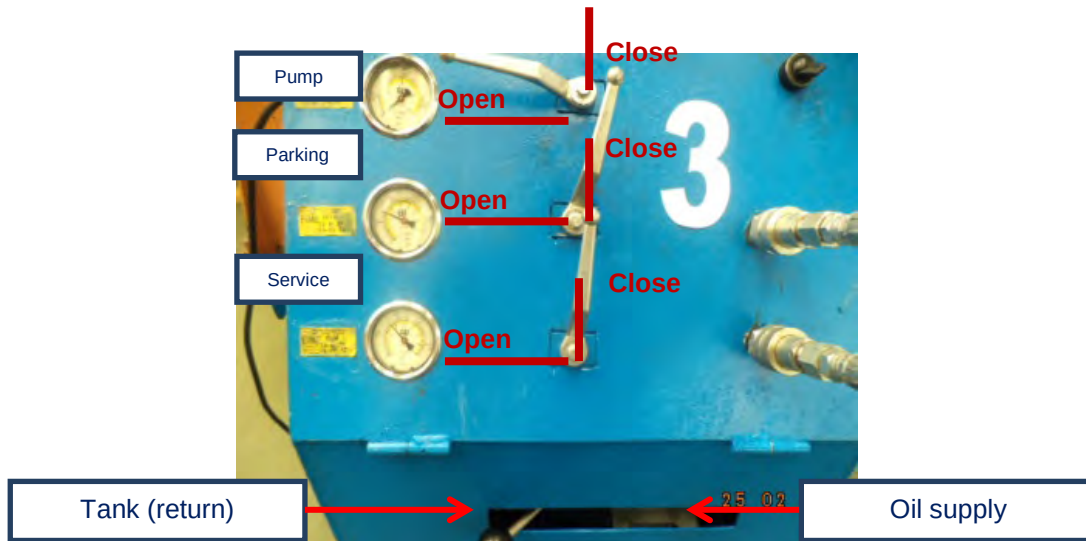
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2.9 Check if the service piston has moved (up) the discs and plates to the applied position as illustrated in the caption.



2.10 Apply and release pressure on the service brake port (steps 2.8 - 2.9) again and repeat this sequence at least 5 times, to ensure proper seating of the piston seals and check for any leaks.

2.11 To check for leakage, leave the service brake pressure applied, close off the service brake valve at the test tooling and then move the oil supply valve to tank return per the caption.



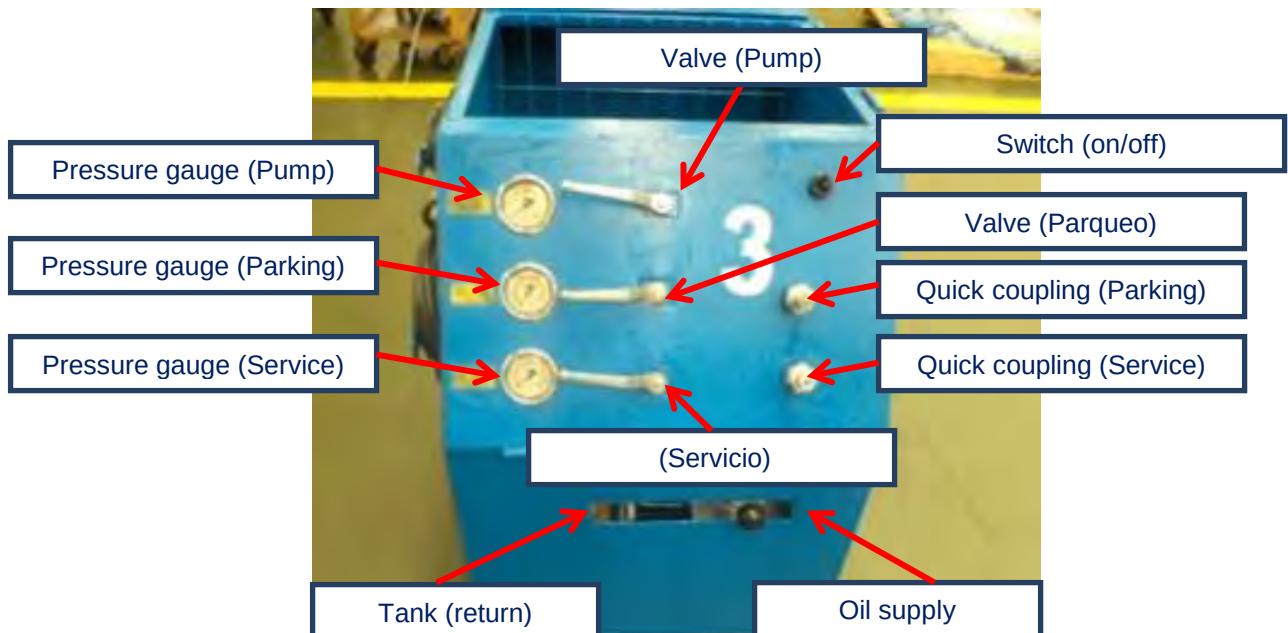
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2.12 Check the parking and service brake pressure for 15 minutes, for any leakage and if no pressure decrease is observed, the brake assembly is considered good.



3.0 Implementation Steps

3.1 Description of parts of the pump



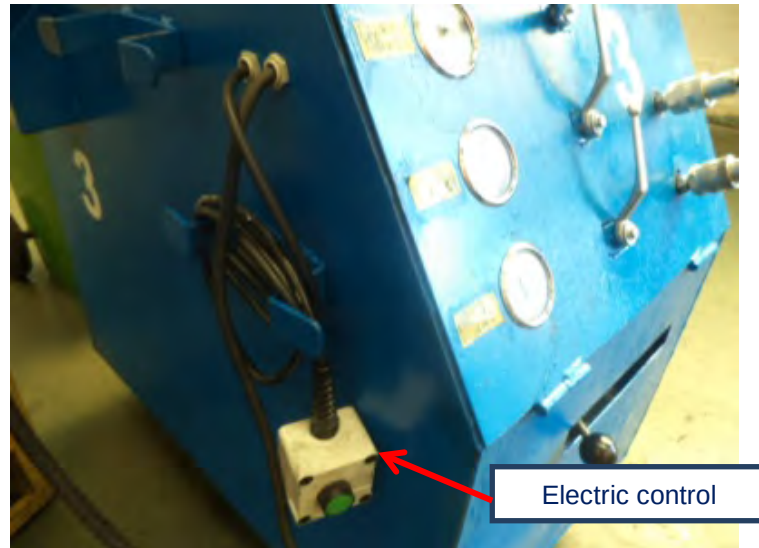
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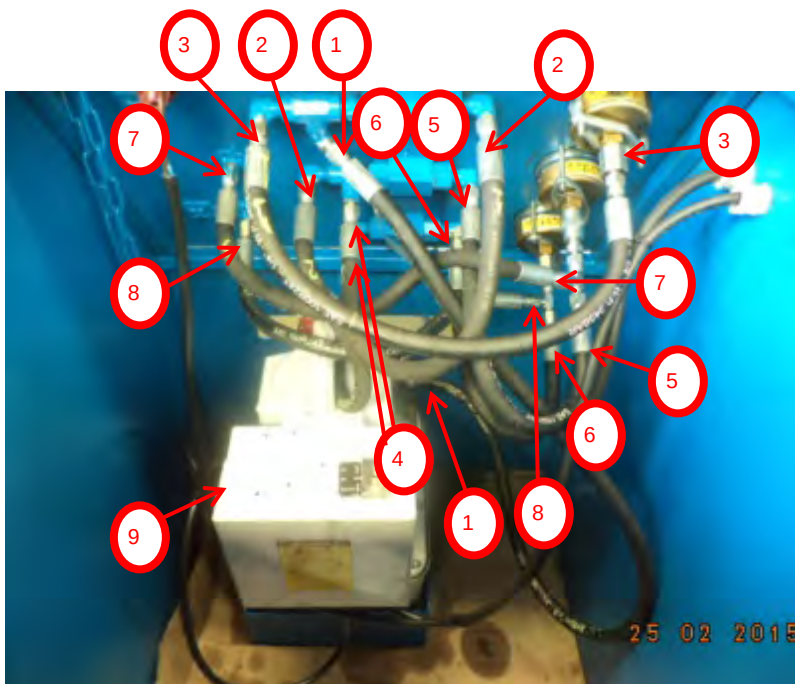
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3.2 Hydraulic pump circuit



- 1.- Main supply hose.
- 2.- Bridge hose (pump - parking).
- 3.- Pressure gauge hose (pump).
- 4.- Bridge hose (Parking-Service).
- 5.- Pressure gauge hose Parking.
- 6.- Pressure gauge hose Service.
- 7.-Taking pressure hose Parking.
- 8.-Taking pressure hose Service.
- 9.- Pump 10.000 PSI.

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3.3 External to the brake hose.



4.0 Benefits

- Quickly identifies any potential leak source.
- Checks application of the park and service brake pistons.
- Avoid unnecessary field repairs and machine downtime.

5.0 Resources Required

- Test Tooling (fabricated)
- Staff training.

6.0 Supporting Attachments / References

- **KENR8374:** 797F and 797F XQ Off-Highway Truck Braking System.
- **RSNR2581:** 797B Camiones de Obras Sistema de frenos.
- **RSNR2611:** 797 Camión de Obras Sistema de frenos.
- **SENR6343:** 793B Off-Highway Truck Air System and Brakes.
- **SSNR1454:** 793C Camión de Obras Sistema de aire y frenos.
- **KENR8738:** 793F and 793F CMD Off-Highway Truck Braking System.
- **RENR8330:** 793D Off-Highway Truck Air System and Brakes.
- **SSNR6331:** 789B Camión de Obras Sistema de Aire y Frenos.
- **KENR9267:** 777F OEM Off-Highway Truck Hydraulic System.
- **RENR3324:** 776D, 777D and 777D HAA Off-Highway Truck/Tractors Air System and Brakes.
- **SSNR2986:** 740 Camión Articulado, 768C, 769C, 769D, 772B, 773B, 773D, 773E, 773F, 776, 776B, 776C, 776D, 777, 777B, 777C, 777D, 777F, 784B, 784C, 785, 785B, 785C, 785D, 789, 789B, 789C, 793, 793B, 793C y 793D de Obras/Tractores, 773E, 773F, 775F, 777F y 785D de Obras/Tractores OEM y 771C, 771D, 775B, 775D, 775E y 775F para Canteras Aux. de tracción automá. electrón. (AETA)

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7.0 Related Best Practices

- None.

8.0 Acknowledgements

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